

## Executive Summary

This project applies a full exploratory data analysis (EDA) and k-means clustering workflow to a large U.S. corporate bankruptcy dataset containing 78,682 firm-year observations from 1999–2018. The goal was to determine whether unsupervised learning could meaningfully separate financially distressed firms from healthy ones using only financial statement variables and engineered ratios. The analysis demonstrates that even without labeled outcomes, financial structure alone contains strong signals that distinguish failed firms from solvent ones.

The dataset includes 21 variables representing assets, liabilities, cash flows, profitability, and derived financial ratios. Initial preprocessing confirmed zero missing values and identified substantial variation in firm size and financial condition. Univariate analysis revealed a highly imbalanced dataset, 93.4% of firms were alive, and 6.6% had failed. Multivariate analysis highlighted strong positive correlations among profitability measures and cash-flow metrics, while leverage-related variables showed distinct patterns associated with distress.

A new engineered feature, the *financial stress index*, was created to capture the combined effects of leverage and earnings capacity. All numeric variables were standardized prior to clustering to ensure equal contribution to distance calculations. A k-means model with  $k = 2$  produced two distinct clusters: one containing nearly all failed firms and another containing primarily solvent firms. Cluster interpretation showed that distressed firms exhibited lower earnings, weaker liquidity, and higher liabilities relative to assets.

Model evaluation—performed by mapping clusters to actual bankruptcy labels—showed strong predictive alignment. The clustering achieved 93.8% accuracy, 93.8% precision, 99.9% recall, and an F1-score of 0.968. The exceptionally high recall indicates that the model

successfully identified nearly all failed firms, a critical requirement in bankruptcy risk analysis where false negatives carry significant financial consequences.

Overall, the project demonstrates that unsupervised learning can uncover meaningful financial patterns associated with corporate distress. The workflow highlights the importance of rigorous EDA, thoughtful feature engineering, and careful interpretation when applying clustering to real-world financial data. These results provide a strong foundation for future work using supervised models, time-series features, and alternative clustering algorithms.